

AN EXPLICIT TAUT-FOLIATION EULER-CLASS OBSTRUCTION ON A HYPERBOLIC MAPPING TORUS

ARGUS OPEN RESEARCH PROJECT

ABSTRACT. We give a completely explicit pair (M_f, w) in the taut-foliation branch of Kirby Problem 3.15. The manifold is the mapping torus of a specified affine pseudo-Anosov automorphism of the genus-three Eierlegende Wollmilchsau origami. In saturated free coordinates,

$$H_1(M_f; \mathbb{Z}) \cong \mathbb{Z}^3 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/6, \quad \text{PD}(w) = (-2, -2, 2).$$

The class w is even, integral, and lies in the dual Thurston unit ball, but it is not the Euler class of any smooth cooriented taut foliation on M_f . The nonrealizability follows from Liu's one-periodic-orbit obstruction after an exact enumeration of all fixed trajectories. All finite algebra, homology, fixed-point, Alexander-polytope, and affine noncollision assertions were independently replayed using exact arithmetic. The contribution is an explicit branch-specific application; it is not a classification of the taut-foliation branch or of the other five branches.

1. INTRODUCTION AND STATEMENT

Let M be a closed oriented hyperbolic three-manifold with $b_1(M) > 0$. Kirby Problem 3.15 asks which real cohomology classes occur as Euler classes of six kinds of structures, the first being taut foliations [1]. Parity and the dual Thurston norm give familiar necessary conditions. They are not sufficient for the taut-foliation branch.

The purpose of this note is to record one exact, externally checkable pair. It combines the quaternion origami of Herrlich–Schmithuesen [3], explicit affine-lift formulas of Matheus–Yoccoz [5], the Alexander–Thurston comparison of McMullen [6], and the obstruction of Liu [4]. The ingredients are established results; the construction below is a specified application with all finite choices made explicit.

Let

$$Q_8 = \{\pm 1, \pm i, \pm j, \pm k\}, \quad i^2 = j^2 = k^2 = ijk = -1.$$

The Eierlegende Wollmilchsau W is the square-tiled surface with one unit square for each $g \in Q_8$, horizontal gluing $g \mapsto gi$, and vertical gluing $g \mapsto gj$. Let $\tilde{S}, \tilde{T}$ denote the standard affine lifts whose derivatives are

$$S = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \quad T = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}.$$

Write L_g for left deck translation by g . Define

$$f = L_{-i} \circ \tilde{T} \circ \tilde{S}^2 \circ \tilde{T} \circ \tilde{S}^2, \tag{1}$$

where the rightmost map is applied first. Equivalently, the affine generators are applied in the order $SSTSSST$, followed by L_{-i} .

Let

$$M_f = W \times [0, 1] / (x, 1) \sim (f(x), 0).$$

The exact homology calculation below supplies a saturated basis (time, u, v) of the free quotient of $H_1(M_f; \mathbb{Z})$. We record a class in $H^2(M_f; \mathbb{Z})$ by the coordinates of its Poincaré dual in this basis.

Theorem 1.1. *The manifold M_f is closed, oriented, and hyperbolic, with*

$$H_1(M_f; \mathbb{Z}) \cong \mathbb{Z}^3 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/6.$$

There is an even integral class $w \in H^2(M_f; \mathbb{Z})$ with

$$\text{PD}(w) = (-2, -2, 2), \quad x^*(w) \leq 1,$$

where x^ is the dual Thurston norm, such that w is not the Euler class of any smooth cooriented taut foliation on M_f .*

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2. THE ORIGAMI MONODROMY

The square decomposition of W has eight faces and sixteen edge classes. The commutator of the horizontal and vertical gluing permutations is the central element -1 , producing four vertex classes. Hence

$$\chi(W) = 8 - 16 + 4 = -4,$$

so W is a closed oriented surface of genus three.

Lemma 2.1. *The derivative of the map in (1) is*

$$A = Df = TS^2TS^2 = \begin{pmatrix} 11 & 4 \\ 8 & 3 \end{pmatrix}.$$

It has characteristic polynomial $X^2 - 14X + 1$ and eigenvalues $7 \pm 4\sqrt{3}$.

Proof. This is direct matrix multiplication. The left deck transformation has derivative the identity, $\det A = 1$, and $\text{tr } A = 14$. $\square$

The derivative is hyperbolic. Since f is an affine automorphism of the translation surface W , it is a pseudo-Anosov standard representative with orientable invariant foliations. Thurston hyperbolization for closed pseudo-Anosov mapping tori therefore makes M_f hyperbolic.

3. INTEGRAL HOMOLOGY AND THE SELECTED CLASS

The cellular chain complex of the eight-square model has sixteen oriented edge generators. Applying the explicit affine lifts to those generators, then imposing the square-boundary and vertex relations, gives the induced integral action on $H_1(W; \mathbb{Z})$. Adjoining the suspension generator and the relations from $f_* - I$ gives a mapping-torus presentation whose Smith factors are

$$1, 1, 2, 6, 0, 0, 0. \quad (2)$$

Consequently

$$H_1(M_f; \mathbb{Z}) \cong \mathbb{Z}^3 \oplus \mathbb{Z}/2 \oplus \mathbb{Z}/6.$$

The exact row and column operations also give three primitive homomorphisms to $\mathbb{Z}$. Their determinant on the free quotient is ± 1 , so the resulting coordinates (time, u , v) are saturated. Define

$$\text{PD}(a) = (1, -1, 1), \quad \text{PD}(e_f) = (-4, 0, 0), \quad w = e_f + 2a. \quad (3)$$

Thus

$$\text{PD}(w) = (-2, -2, 2) = 2(-1, -1, 1).$$

In particular, w is an even integral class, not merely a rational point of $H^2(M_f; \mathbb{R})$. The time coordinate gives

$$\langle a, [W] \rangle = 1. \quad (4)$$

4. THE DUAL THURSTON CONDITION

Let $P \subset \mathbb{R}^3$ be the Newton polytope of the multivariable Alexander polynomial in the saturated free coordinates. Its support has unique bottom and top vertices in the fiber direction,

$$p_- = (0, 2, 2), \quad p_+ = (4, 2, 2).$$

McMullen's equality on the fibered cone identifies the canonical Euler functional with

$$\text{PD}(e_f) = p_- - p_+ = (-4, 0, 0),$$

in agreement with (3).

The dual Alexander unit ball is the difference body $P - P$ in the normalization used here. Two support points satisfy

$$(-2, -2, 2) = (0, 2, 2) - (2, 4, 0), \quad (5)$$

so $\text{PD}(w) \in P - P$. An independent facet computation gives twelve primitive supporting inequalities; direct substitution places w on the facet $-x - y = 4$. It also checks every one of the 137 distinct support differences against all twelve inequalities.

Because $b_1(M_f) = 3$, McMullen's global comparison

$$\|\phi\|_A \leq \|\phi\|_{\text{Th}}$$

applies. Duality reverses the inclusion, giving

$$B_A^* \subseteq B_{\text{Th}}^*.$$

Together with (5), this proves

$$x^*(w) \leq 1. \quad (6)$$

No equality assertion for $x^*(w)$ is needed.

5. COMPLETE EXCLUSION OF ONE-PERIODIC TRAJECTORIES

The four singularities of the Abelian differential are permuted by f as two transpositions. Thus no singularity is fixed. On the base torus,

$$|\det(A - I)| = 12,$$

so there are exactly twelve base fixed points. Exhausting all eight Q_8 -lifts over each base point yields precisely twelve regular fixed points of f . Their suspension homology classes are as follows.

TABLE 1. Complete one-periodic suspension multiset.

Free homology class	Multiplicity
$(1, -1, 0)$	1
$(1, -1, 2)$	1
$(1, 0, 0)$	8
$(1, 1, -2)$	1
$(1, 1, 0)$	1

For completeness, the enumeration proceeds by choosing a rational based path to each fixed lift, applying the explicit free-group automorphism induced by f , Schreier-rewriting the resulting closed word in the eight-square cover, and reducing it in the saturated mapping-torus basis. The sole fixed point lying on a square edge is approached through a fixed rational interior waypoint, avoiding an unrecorded choice of edge sector.

Every regular fixed point has local index -1 . Their total index is therefore -12 , which agrees with the Lefschetz number

$$L(f) = 2 - \text{tr}(f_*|H_1(W; \mathbb{Z})) = -12.$$

As a separate check, the first fiber layer of the exact multivariable Alexander polynomial encodes exactly the five classes and coefficients in Table 1. The enumeration is therefore complete, rather than a bounded orbit search.

By (3),

$$\text{PD}(a) = (1, -1, 1),$$

which does not occur in Table 1. Hence no one-periodic trajectory represents $\text{PD}(a)$.

6. LIU'S OBSTRUCTION AND PROOF OF THE THEOREM

We use the following consequence of Liu's one-periodic-orbit criterion [4, Lemma 3.2]. Let F be a closed oriented surface and let $g : F \rightarrow F$ be a pseudo-Anosov standard representative. If an integral class c on the mapping torus satisfies

$$\langle c, [F] \rangle = 1$$

and $\text{PD}(c)$ is not represented by a one-periodic trajectory, then the class $e_g + 2c$ is not the real Euler class of any weakly symplectically fillable contact structure.

Proof of Theorem 1.1. The closedness, orientation, hyperbolicity, and homology statement follow from Lemma 2.1 and (2). The saturated coordinate calculation proves that w is even and integral, and (6) proves the required dual Thurston bound.

Apply Liu's criterion with $F = W$, $g = f$, and the class a in (3). The fiber evaluation is one by (4), while Table 1 proves that $\text{PD}(a)$ has no one-periodic representative. Therefore

$$w = e_f + 2a$$

is not the Euler class of any weakly symplectically fillable contact structure.

The Eliashberg–Thurston approximation theorem, in the form recalled and used by Liu, perturbs any smooth cooriented taut foliation to a weakly fillable contact structure without changing the Euler class of the oriented

plane field [2, 4]. A smooth cooriented taut foliation with Euler class w would therefore contradict the preceding obstruction. $\square$

7. EXACT COMPUTATIONAL CERTIFICATION

The construction involves substantial but finite algebra. The accompanying certificate was produced and replayed using integer and rational arithmetic; no floating-point inequality is used in the proof of Theorem 1.1. The independent replay checks the following items.

- (1) The full multiplication table of Q_8 , the eight-square cellular boundary maps, and the canonical affine actions of $\tilde{S}$ and $\tilde{T}$.
- (2) The derivative word $SSTSST$, the deck postcomposition $-i$, the integral chain action, all Smith factors in (2), and saturation of the three free coordinates.
- (3) All base fixed points, all Q_8 -lifts, the four singularity images, every based Schreier word, and every homology class in Table 1.
- (4) The Alexander support, its first fiber layer, all 137 support differences, the complete rational facet system, and the membership of w in the difference body.
- (5) Every hypothesis used in the application of Liu’s criterion.

There is a close published affine formula in Matheus–Yoccoz [5, Remark 2.9]. It does not give the same pair. Conjugating the derivative of f by S^2 gives

$$B = \begin{pmatrix} 3 & 4 \\ 8 & 11 \end{pmatrix},$$

whereas the printed element has derivative $-B$. Thus the traces are 14 and -14 , respectively. An exact simultaneous intertwiner system for conjugacy and inverse conjugacy has rank eight and nullity zero. Moreover, the same-derivative factor in the published formula acts trivially on the normal deck group Q_8 , while the present map induces the nontrivial automorphism Ad_i . The corresponding mapping tori have first Betti numbers three and five. These independent exact discriminators exclude the nearest identified collision.

8. SCOPE AND ORIGINALITY BOUNDARY

Theorem 1.1 gives one explicit nonrealizable Euler class in the smooth cooriented taut-foliation branch of Kirby Problem 3.15. It does not classify all Euler classes in that branch. It makes no assertion about tight contact structures, universally tight contact structures, pseudo-Anosov flows, quasigeodesic flows, or circle actions.

The obstruction theorem of Liu, the quaternion origami, its deck group, the canonical affine lifts, McMullen’s norm comparison, and the taut-foliation approximation theorem are prior work. The project contribution is the exact selection and certification of the pair (M_f, w) , including the complete one-periodic enumeration and the exact rejection of the nearest published affine formula.

A targeted audit of the cited primary sources, source archives, scholarly indexes, and public mathematical repositories found no equivalent explicit pair under affine conjugacy, deck relabelling, monodromy inversion, and simultaneous orientation/coorientation reversal. This supports the limited description “previously unlocated explicit application within the audited corpus.” It is not a claim of worldwide priority.

9. INDEPENDENT-REFEREE CHECKLIST

An external referee can audit the result in the following order.

- (1) Verify the eight-square Q_8 gluing and the genus-three Euler characteristic.
- (2) Multiply the six derivative generators and check $Df = \begin{pmatrix} 11 & 4 \\ 8 & 3 \end{pmatrix}$.
- (3) Recompute the mapping-torus Smith factors and confirm that the displayed free coordinates are saturated.
- (4) Check the twelve base fixed points, all lifted fixed points, and the complete homology multiset in Table 1.
- (5) Recompute the Alexander support and verify both $\text{PD}(e_f) = (-4, 0, 0)$ and $\text{PD}(w) \in P - P$.
- (6) Check the hypotheses of Liu’s Lemma 3.2 and the passage from taut foliations to weakly fillable contact structures.
- (7) Keep the conclusion restricted to one explicit taut-foliation class, and keep the originality statement restricted to the audited corpus.

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